

EXPERIMENT 3

Water Jug Problem

Aim

To write a Python program to solve the Water Jug Problem using the State Space Search approach.

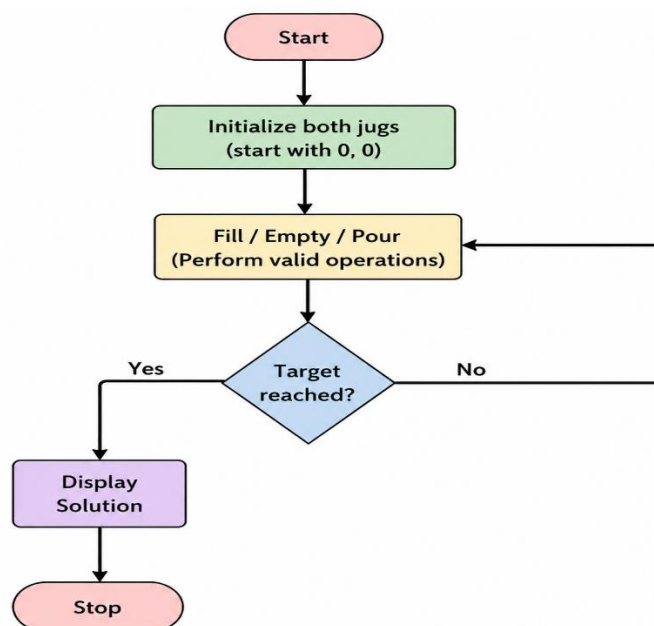
Objective

To find the sequence of steps required to obtain the desired quantity of water using two jugs with different capacities.

Algorithm

1. Start with both jugs empty.
2. Fill one jug completely.
3. Pour water from one jug to another until one becomes empty or the other becomes full.
4. Empty a jug whenever necessary.
5. Repeat the above steps until the target quantity is obtained.
6. Display the sequence of states.

Flowchart



Python code:

```
from collections import deque
```

```
def water_jug(jug1, jug2, target):
```

```
    visited = set()
```

```
    queue = deque([((0, 0), [])])
```

```
    while queue:
```

```
        (a, b), path = queue.popleft()
```

```
        if (a, b) in visited:
```

```
            continue
```

```
        visited.add((a, b))
```

```
        path = path + [(a, b)]
```

```
        if a == target or b == target:
```

```
            return path
```

```
    next_states = [
```

```
        (jug1, b),
```

```
        (a, jug2),
```

```
        (0, b),
```

```
        (a, 0),
```

```
        (max(0, a-(jug2-b)), min(jug2, a+b)),
```

```
        (min(jug1, a+b), max(0, b-(jug1-a)))
```

```
    ]
```

```
    for state in next_states:
```

```
        if state not in visited:
```

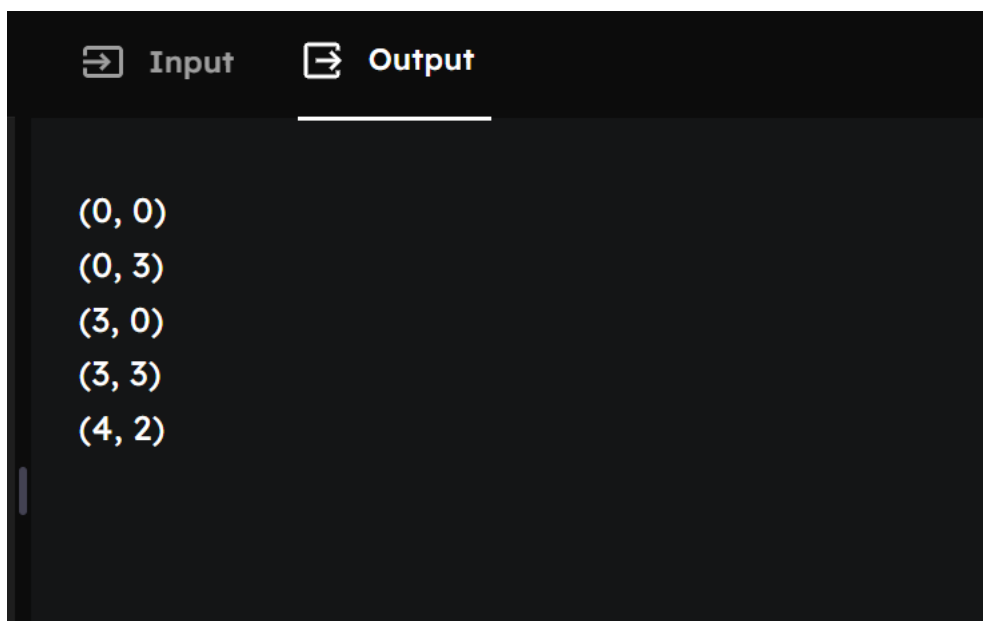
```
        queue.append((state, path))

    return None

solution = water_jug(4,3,2)

for step in solution:
    print(step)
```

Output:

A screenshot of a code editor with a dark theme. At the top, there are two tabs: 'Input' and 'Output'. The 'Output' tab is selected and underlined. Below the tabs, the output of the program is displayed as a list of tuples: (0, 0), (0, 3), (3, 0), (3, 3), and (4, 2).

```
Input  Output
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
```

Result

The Water Jug Problem was solved successfully using Python.

Conclusion

The program demonstrates state space search to obtain the required quantity of water efficiently.